Definitions and Values

Name: _____ Date: _____

As a reminder, these are some constructors provided by the book:

`Circle(int radius, String color)` `Rect(int width, int height, String color)`

Q1. Does each line run? If not, say what is wrong in a few words.

definition	runs?	if not, what is wrong
`Pict box = new Rect(20, 30, "blue");`		
`string name = "Val";`		
`Pict p = Circle(25, "red");`		
`int count = "5";`		
`Pict dot = new Circle(25);`		
`Pict p = new Circle("red", 25);`		

Q2. For each of the two definitions below, label all the parts you can with **type**, **name**, **value**, **constructor**, **argument**

```
Pict dot = new Circle(25, "red");
```

```
int year = 2026;
```

Q3. Write a definition for a green rectangle 10 wide and 60 tall, named `stem`.

Q4. Write the value each name holds after this program runs (it doesn't have errors)

```
int total = 7 * 3;
int half = total / 2;
String label = "Total: " + total;
int agePlusOne = 17 + 1;
String textPlusOne = "17" + 1;
```

`total` = _____

`half` = _____

`label` = _____

`agePlusOne` = _____

`textPlusOne` = _____

Q5. Two students average the scores 7 and 8. Neither line is an error.

```
int mean1 = (7 + 8) / 2;
```

```
double mean2 = (7 + 8) / 2.0;
```

`mean1` = _____ `mean2` = _____ Which belongs in a grade report? _____

Q6. In "Does each line run", Java would report each of the errors and refuse to run those programs. Would you have rather gotten an error from the `mean` example? Why or why not?